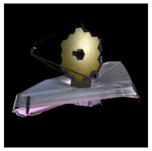

9897 - How does a galaxy blow out bubbles? --- the launching site of a nuclear outflow

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
JWST NIRSpec IFU observations of NGC3079				
	1		NIRSpec IFU Spectroscopy	(1) NGC3079
	2		NIRSpec IFU Spectroscopy	(2) NGC3079-BACKGROUND

ABSTRACT

Galaxies could blow out superbubbles with either nuclear starburst or AGN. Studying these nuclear superbubbles is important to understand how galaxies give feedback to their environments and suppress SF. Despite extensive studies of various large-scale galactic outflow structures, we still lack an in-depth view of the galactic nuclear region responsible for the launch of the outflow, largely limited by the sensitivity and angular resolution of some IR lines that are free from the heavy extinction.

We propose a JWST/NIRSpec IFU observation of the best example of a galactic nuclear superbubble in NGC3079. The galaxy hosts one of the best defined bi-polar kpc-scale superbubbles, believed to still be in its early evolutionary stage, and is an imprint of the interplay between various feedback sources and mechanisms. These energetic processes could also be responsible for the ~60kpc superwind, which is the largest in the local universe.

With the proposed JWST observations, we will conduct spatially resolved spectroscopy of a few emission lines tracing both the ionized and warm molecular gases, and compare the results to multi-wavelength data and numerical simulations. We will explore some scientific problems, including estimating some key parameters of galactic feedback, quantifying the superbubble energy budget, determining the driving force of outflow, examining the coupling of different gas phases, constraining the turbulence in the ISM, measuring the mass and kinetic energy of the molecular outflow, and studying its excitation mechanisms. These studies will provide an in-depth view of how the outflows are launched in a volume much smaller than where they have impacted.

OBSERVING DESCRIPTION

We propose JWST/NIRSpec IFU observations of the nucleus of NGC3079, which is expected to be the launching site of the bi-polar kpc-scale superbubbles, as well as some gaseous structures extending to ~60kpc above the galactic disk. We choose the G235H/F170LP grating-filter combination, which could reach a spectral resolution of R~2,700 and cover a few important emission lines such as the Paalpha, Brgamma, HeI, and H2. These lines will allow us to study the properties and dynamics of different gas phases. We choose a 4-Point-Dither pattern to reach a balance between a well spatial sampling of the PSF, the mitigation of some detector artifacts, the sensitivity, and the overhead. We also request to observe a dedicated background area in a non-interruptive group observation with the source area.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC3079	RA: 10 01 57.9918 (150.4916325d) Dec: +55 40 47.68 (55.67991d) Equinox: J2000	Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Field galaxies, Radio lobes, Starburst galaxies] Extended=YES				
(2)	NGC3079-BACKGROUND	RA: 10 02 6.0000 (150.5250000d) Dec: +55 46 58.00 (55.78278d) Equinox: J2000		
Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Field galaxies, Radio lobes, Starburst galaxies] Extended=YES				

Observation

Proposal 9897, Observation 1

Fri Mar 13 18:01:25 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observations:[Observation 2]

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC3079	RA: 10 01 57.9918 (150.4916325d) Dec: +55 40 47.68 (55.67991d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Active galactic nuclei, Active galaxies, Field galaxies, Radio lobes, Starburst galaxies]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G235H/F170LP	NRSIRS2	20	4	false	true	NONE	4	16	23575.646	

Special Requirements

Aperture PA Range 292.97164917 to 293.97164917 Degrees (V3 154.0 to 155.0)

Sequence Observations 1, 2, Non-interruptible

Observation

Proposal 9897, Observation 2

Fri Mar 13 18:01:25 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observation For: [Observation 1]

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	NGC3079-BACKGROUND	RA: 10 02 6.0000 (150.5250000d) Dec: +55 46 58.00 (55.78278d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Active galactic nuclei, Active galaxies, Field galaxies, Radio lobes, Starburst galaxies]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G235H/F170LP	NRSIRS2	20	1	false	true	NONE	4	4	5893.912	

Special Requirements

Sequence Observations 1, 2, Non-interruptible